

QUADRATFREI (SQUARE-FREE): In mathematics, of an integer, containing no repeated prime factors. Examples include 1, 2, 3, 5, 6, 7, 10, 11, 13, 14 and 15.

QUADRATIC (QUADRATIC FORM): 1. Of the second degree. 2. Of an expression, function or equation, containing one or more terms in which the variable is raised to the second power, but no variables are raised to a higher positive power. Examples are $3x^2$, $a^2 + 2ab + b^2$ and $x^2 + x - 30$.

QUADRATIC CONGRUENCE: A congruence of the form $f(x) \equiv 0 \pmod{m}$, where f is a quadratic polynomial with integer coefficients.

QUADRATIC EQUATION: An equation containing one or more terms raised to the power two; or a polynomial equation of second degree. Its highest exponent or power is two and it is of the form $ax^2 + bx + c = 0$, where a , b and c are given real numbers, with $a \neq 0$. Quadratic equation can be solved by factorisation, completing the squares or by using the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The expression

$b^2 - 4ac$ is referred to as the discriminant. If $b^2 - 4ac > 0$, the equation has two real roots; when $b^2 - 4ac < 0$, then there are two complex roots; and when $b^2 - 4ac = 0$, then there are two equal roots and for two real roots, $b^2 - 4ac \geq 0$.

QUADRATIC FORMULA: The quadratic formula says that the solution for x in the equation is $ax^2 + bx + c = 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

QUADRATIC FUNCTION: A function of the form $f: x \rightarrow ax^2 + bx + c$, where a , b and c are real numbers and $a \neq 0$. The quadratic term is ax^2 with a as its coefficient. The term bx is the linear term with b as its coefficient. The term c is the constant term. The graph of a quadratic function is a parabola, with its axis parallel to the y -axis. If $a > 0$, the curve opens upwards with the lowest point

on it being the vertex, with a point minimum at $x = \frac{-b}{2a}$. The y -value at the vertex is the minimum value of y . If $a < 0$, the curve opens downwards with the highest point on it, the vertex, a maximum point. The y -value at the vertex is the maximum value of y at $x = \frac{-b}{2a}$. The vertex is a turning point on the parabola. If $|a|$ increases, the parabola becomes narrower; as $|a|$ decreases, the parabola becomes wider.

QUADRATIC GRAPH: A graph whose general equation is given by $y = ax^2 + bx + c$, where a , b and c are constants, $a \neq 0$. When a is positive ($a > 0$), the graph is concave upwards and the turning point or vertex of the graph is the lowest or minimum point of the curve which occurs at $x = -\frac{b}{2a}$. The value of y at this point is the minimum value of the function. When a is negative (i.e. $a < 0$) the graph opens downwards and the turning point or vertex is highest, greatest or maximum point on the graph. There is only one turning point (stationary point) in a quadratic graph. This can either be minimum or maximum.

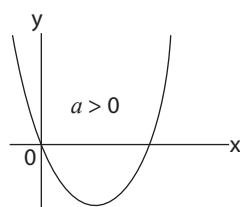


Fig. I: Concave Upwards

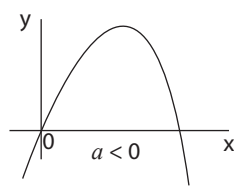


Fig. II: Concave Downwards

Quadratic Graph

QUADRATIC MEAN: The measure of central tendency which is

$$\text{given by } Q = \sqrt{\frac{X_1^2 + X_2^2 + X_3^2 + \dots + X_n^2}{n}} = \sqrt{\frac{\sum_{i=1}^n X_i^2}{n}}$$

QUADRATIC PROGRAMMING PROBLEM: A problem in mathematical programming in which the objective function is a quadratic and the constraints are linear.

QUADRATIC RECIPROCITY (GAUSSIAN RECIPROCITY): The number-theoretic result that if p and q are distinct odd prime numbers, p is a quadratic residue of q if and only if q is a quadratic residue of p , unless both are congruent to 3 modulo 4, in which case the opposite holds, in one case, p is a quadratic residue of q if and only if q is not a quadratic.

QUADRATIC RESIDUE: A number congruent to a perfect square to a given modulus; a is a quadratic residue (mod n) if and only if $x^2 \equiv a \pmod{n}$ is soluble for integral x . When n is prime, this holds precisely when the Legendre symbol, $(a|n)$, is unity.

QUADRATIC SEQUENCE: An expression in which the changes in the number sequence follow a certain uniform pattern, for example in the following sequence: 1, 4, 9, 16..., shows there is a pattern, which is +3, +5 and +7. If the formula $n^2 + n$ is used to generate a sequence, it means that when $n = 1$, $1^2 + 1 = 2$; when $n = 2$, $2^2 + 2 = 6$; when $n = 3$, $3^2 + 3 = 12$; when $n = 4$, $4^2 + 4 = 20$, giving the sequence 2, 6, 12, 20. The following rules apply to quadratic sequences: If the second difference is 2, start with n^2 . If the second difference is 4, start with $2n^2$. If the second difference is 6, start with $3n^2$.

QUADRATICS: The branch of algebra responsible for the studies of quadratic equations.

QUADRATURE: 1. The position of a celestial body such as the Moon or any other planet where a line between it and Earth or another celestial body makes a right angle with a line joining from a third. 2. In mathematics, it is the process of constructing a square with the same area as a given figure or surface, especially a circle. 3. **Numerical quadrature** is the evaluation of a definite integral by a formula involving weighted sums of function values at given points. The rule is said to have order n if it is exact for polynomials of that degree. The trapezoidal rule has order 1 while Simpson's rule has order 3.

QUADRATURE AMPLITUDE MODULATION (QAM): A widely used modulation technique that combines two amplitude-modulated signals into a single channel. Two carriers of the same frequency by different phases are used, with a phase difference of 90° .

QUADRI-: A prefix used to denote four.

QUADRIC: 1. Characterised by an algebraic equation of the second degree, usually in two or three variables. 2. Of a curve, function or surface in three dimensions, described by equations containing the squares of x , y and z , but no higher powers of them. Examples of such surfaces include the sphere, the ellipsoid, the cone and the cylinder.

QUADRIC CONE: A quadric surface that is given by the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}, \text{ using a particular coordinate system. If a cross-section}$$

is taken parallel to the xy -plane an ellipse is formed. In the case where $a = b$ a circle is formed. Cross-sections parallel to the planes of the other axes result in hyperbolas.

QUADRIC CURVE: The graph of a second degree equation in two variables.

QUADRIC SURFACE: The graph of a second degree equation in three variables.

QUADRICEPS: A group of extensor muscles, namely rectus femoris, vastus intermedius, vastus lateralis and vastus medialis that cover the anterior surface and sides of the mammalian femur. The rectus femoris originates in the pelvis but the others originate from different points

on the thigh bone and join in a single tendon which crosses the knee cap to insert onto the shin bone. They are usually the most powerful muscles in the body and are the major extensors for straightening the leg. They play a vital role in any activity that involves running, walking, jumping or even standing.

QUADRIFOLIATE: Having four leaves or leaflets.

QUADRILATERAL: A two dimensional figure with four sides. Examples include a square, rectangle, rhombus, parallelogram and trapezium.

QUADRINOMIAL: An algebraic expression consisting of four terms connected by the signs plus (+) or minus (-).

QUADRIVALENT (TETRAVALENT): An element such as carbon and silicon with a valency of four.

QUADRO-: An affix used in names to denote four atoms bound into a quadrangle, for example, a square.

QUADRUPEDAL: It describes an animal that walks on four feet.

QUADRUPOLE: A set of four point charges or two negative and two positive charges arranged on the edges of a square that has zero monopole moment or net charge and dipole moment of zero, when the coordinate origin is at the centre of the square. An example is a carbon dioxide (CO_2) molecule.

QUADRUPOLE ION STORAGE TRAP (QUISTOR): An arrangement in which ions with a desired range of quotients mass/charge are trapped by making them describe stable paths under the effect of a static and a high-frequency electric quadrupole field.

QUADRUPOLE MASS ANALYSER: An arrangement in which ions with a desired quotient mass/charge are made to describe a stable path under the effect of a static and a high-frequency electric quadrupole field and are then detected. Ions with a different mass/charge are separated from the detected ions because of their unstable paths.

QUALITATIVE: Referring to descriptions or distinctions based on the quality or character of an element, compound, etc., as opposed to its size or quantity.

QUALITATIVE ANALYSIS: Analysis in which substances are identified or classified on the basis of their chemical or physical properties, such as chemical reactivity, solubility, molecular weight, melting point, radiative properties (emission, absorption), mass spectra and nuclear half-life.

QUALITATIVE DATA (CATEGORICAL DATA): Data that can be arranged into categories and describes characteristics such as colours, smells, texture, etc. It does not have a number associated with it.

QUALITATIVE ELEMENTAL SPECIFICITY: Ability of a method to detect one element in the presence of another element.

QUALITATIVE TEST: A test on a sample of a compound or element, which gives observations indicating the presence of that element or compound. It may involve the separation and identification of the ions, for example, in an aqueous solution.

QUALITY ASSURANCE: Any technique or method for collecting, processing or analysing data with the aim of maintaining or improving the reliability or validity of the data.

QUALITY CONTROL: The analysis of a sequence of small samples taken at regular intervals from the output of an industrial production process with the aim of ensuring that the output meets its required specification.

QUALITY FACTOR (RELATIVE BIOLOGICAL EFFECTIVENESS):

The linear-energy-transfer dependent factor by which absorbed dose is multiplied to obtain dose equivalent, or constant of proportionality between dose equivalent and absorbed dose.

QUALITY OF SERVICE (QOS): A measure of the transmitting performance of a communication system transmission rate, error rate, etc.

QUALITY OF SOLVENT: A qualitative characterisation of the polymer-solvent interaction. A solution of a polymer in a 'better' solvent is characterised by a higher value of the second virial coefficient than a solution of the same polymer in a 'poorer' solvent.

QUANTIC: In mathematics, it is a homogeneous function of two or more variables, in a form with rational and integral coefficients, especially a quadratic form. For example, $x^2 + 3xyz + y^2$ is a **ternary quadric quantic**.

QUANTIFIER: 1. In logic, a symbol of predicate calculus that contains a variable and indicates the generality of the open sentence in which that variable occurs. The symbol for the universal quantifier "all" or "for all" is " $\forall$ " and for the existential quantifier "exists" or "there exists" is " $\exists$ ". 2. Any analogous symbol in an extended logic, such as $(Mx)Fx$ for 'most x are F ' or restricted quantifiers such as $(x: Fx) Gx$ for 'all F s are G s'.

QUANTIFY: 1. In mathematics, to determine, indicate or express a quantity. 2. In logic, to use a quantifier to bind a variable; or to determine, indicate or make explicit the quantity of a proposition.

QUANTILES: A term used for the division of a probability distribution into equal ordered subgroups. Examples include decile, percentile and quartile.

QUANTITATIVE ANALYSIS: Analyses in which the amount or concentration of an analyte may be determined (estimated) and expressed as a numerical value in appropriate units. Qualitative analysis may take place without quantitative analysis, but quantitative analysis requires the identification (qualification) of the analytes for which numerical estimates are given.

QUANTITATIVE DATA: Numerical data such as length, time and temperature which can be measured.

QUANTITATIVE INHERITANCE (MULTIFACTORIAL INHERITANCE; POLYGENIC INHERITANCE): The determination of a phenotypic characteristic or trait, for example, height, skin colour, intelligence and obesity caused by many genes (quantitative trait loci) or the interaction with the environment or both. Characteristics controlled in this way show continuous variation. Diseases, which have some genetic traits, are polygenic. Examples are autism, cancer, diabetes, etc.

QUANTITATIVE STRUCTURE-ACTIVITY RELATIONSHIP (QSAR): A process that quantitatively defines the relationship between the chemical structure of drug and its effect on an organism. It is often used to predict the activity or toxicity of new drugs. It can also be used, for example, to express biological activity quantitatively such as in the concentration of a substance required to give a certain biological response.

QUANTITATIVE TEST: The measurements such as mass and volume of the amount of substances present in a sample.

QUANTITATIVE TRAIT LOCUS (QTL): A region of a chromosome containing one or more genes that contributes to determine a quantitative trait. Typically, several of such genes called polygenes, each having a small effect, act together to specify a trait. QTLs can be molecularly identified, for example, with Amplified Fragment Length

Polymorphism (AFLP), to help map regions of the genome that contain genes involved in specifying a quantitative trait. This can be an early step in identifying and sequencing these genes.

QUANTITATIVE VARIABLE: A variable with numerical modalities. Examples are weight, dimension, age, speed and time. Discrete variables such as number of children per family can be distinguished from continuous variables such as length of a jump.

QUANTITY: 1. A mathematical entity that has a numerical value or magnitude, for example, five apples. 2. A numerical expression; a variable that ranges over numbers. 3. In logic, the property of being either universal or particular. 4. The product of a measurable phenomenon such as electric current or radiation intensity and the time during which the phenomenon is measured.

QUANTITY CALCULUS: Algebra with quantities where the symbols of quantities represent products of numerical values and their units.

QUANTITY OF DIMENSION ONE (DIMENSIONLESS QUANTITY): Quantity in the dimensional expression of which all the exponents of the dimensions of the base quantities are zero.

QUANTISATION: The process of constructing a quantum field theory for a system, from the original classical field theory. The formulation of the quantum theory for the system can be performed using formalisms such as matrix mechanics or wave mechanics. The application of these methods leads to the conclusion that energy level in systems, such as atoms, are discrete or have particular values rather than continuous.

QUANTISE: 1. In mathematics, to restrict a quantity or variable to values that are integral multiples of some unit. 2. In physics, to change the description of (a physical system) from classical to quantum-mechanical, usually resulting in discrete values for observable quantities, as energy or angular momentum.

QUANTUM: The minimum amount by which certain properties of a system, such as energy or mass, electromagnetic radiation or angular momentum, associated with a physical system, such as an atom, molecule or elementary particle can change in a wave or field; or the smallest discrete quantity of a physical property; or the smallest unit used to measure a physical property. For example, the quantum of electromagnetic radiation is the photon.

QUANTUM CHAOS: A branch of physics which studies how chaotic classical reactions can be described in terms of quantum theory. Quantum chaos describes and tries to understand the nature of the wave-like motions for the electrons in atoms and molecules (quantum mechanics), as well as electromagnetic waves and acoustics, etc. Systems exhibiting chaos in classical mechanics appear not to necessarily exhibit chaos in quantum mechanics.

QUANTUM CHEMISTRY: The application of quantum mechanical principles to chemical substances in determining physical and chemical properties from empirical methods and first principles, for example, the electronic behaviour of atoms and molecules as pertaining to their reactivity.

QUANTUM CHROMODYNAMICS (QCD): A quantum field theory of elementary particles that describes the properties of the strong interactions between quarks and between protons and neutrons in the framework of quantum theory.

QUANTUM COMPUTING: Computer systems under development based on quantum mechanics that use components at the subatomic, atomic or molecular level, to store or process information. It uses the quantum bit (qubit) to encode information, which can hold an infinite number of values; making them much faster and capable of performing more complex calculations than present day computers.

QUANTUM COUNTER: A medium emitting with a quantum yield independent of the excitation energy over a defined spectral range, for example, concentrated rhodamine 6G solutions between 300 and 600 nm. Also used for devices producing an electrical signal proportional to the photon flux absorbed in a medium.

QUANTUM CRYPTOGRAPHY: Cryptography that is based on quantum mechanics that if a message exists as a quantum state, then any eavesdropper would change the state and, therefore, can be detected. Examples of quantum cryptography are the use of quantum communication to securely exchange a key called **quantum key distribution (QKD)** and the hypothetical use of quantum computers that would allow the breaking of various popular public-key encryption and signature schemes. **Quantum key distribution (QKD)** uses quantum mechanics to guarantee secure communication. It enables two parties to produce a shared random secret key known only to them, which can then be used to encrypt and decrypt messages.

QUANTUM DIMENSIONS: As yet undiscovered dimensions of space-time that convert force particles to matter particles and vice versa. Such dimensions are predicted in theories with supersymmetry or superstrings. The pair of particles that are converted into each other are called **superpartners**.

QUANTUM DOT: A quantum-mechanical system, usually made from a semiconductor, in which electrons can be confined into a small region a few nanometres in size containing a few thousand atoms and have electronic properties intermediate between those of bulk semiconductors and those of discrete molecules. Such systems act as artificial atoms with their own set of quantum states for the electrons. They were discovered in the 1980s by Alexei Ekimov, a Russian solid state physicist in a glass matrix and by Louis E. Brus, a professor of chemistry at Columbia University in colloidal solutions.

QUANTUM ELECTRODYNAMICS (QED): The theory describing the interaction of electromagnetic radiation with electrically charged matter within the framework of relativity and quantum mechanics.

QUANTUM ELECTRONICS: The branch of electronics associated with the various energy states of matter, motions within atoms or groups of atoms and various phenomena in crystals and its application in the design of devices such as the atomic hydrogen maser and the caesium atomic-beam resonator.

QUANTUM ENTANGLEMENT: The phenomenon in quantum mechanics in which a particle or system does not have a definite state but exists as an intermediate form of two states. Measuring any one of the particles' quantum state determines the quantum state of the other particle.

QUANTUM ERROR CORRECTION: A technique used in quantum computing to prevent errors due to phenomena like decoherence and other quantum noise being propagated.

QUANTUM FIELD THEORY: A theory in quantum mechanics that applies to systems that have an infinite number of degrees of freedom. Examples include: (i) the **quantum electrodynamics (QED)** which is applied to the understanding of electromagnetism in which the photon is emitted or absorbed by particles, the photon being the quantum of the electromagnetic field; and (ii) **quantum chromodynamics (QCD)** applied to the understanding of the strong interactions between quarks and between protons, neutrons and other baryons and mesons.

QUANTUM FLUCTUATIONS (VACUUM FLUCTUATIONS): A short-lived change in the amount of energy in a point in space due to the Heisenberg uncertainty principle. Quantum fluctuations can often involve the rapid production and annihilation of virtual pairs of particles and their antimatter counterparts.

QUANTUM GRAVITY (QG): An aspect of quantum theory that attempts to incorporate the gravitational field described by the theory of relativity. The theory must be able to predict the outcome of situations where both quantum effects and strong-field gravity are important. An important aspect of quantum field theory in curved space-time is its description of the 'Hawking Processes'. Black holes can also be interpreted as requiring a quantum theory of gravity.

QUANTUM GROUP: A mathematical structure that is associated with the symmetry of noncommutative geometry in the same way that the Lorentz group is associated with Minkowski space-time in the special theory of relativity.

QUANTUM HALL EFFECT: A phenomenon exhibited by certain semiconductor devices at low temperatures and high magnetic fields, in which the Hall coefficient R_H is proportional to h/e^2 , where h is the Planck constant and e is the charge of the electron.

QUANTUM HYPOTHESIS: 1. The hypothesis that nerve impulses are transmitted across junctions (synapses) in the nervous system by the release of discrete packets or quanta or neurotransmitter, each quantum representing the contents of single synaptic vesicle. 2. A hypothesis that some physical quantity can assume only a certain discrete set of values; examples are Planck's law and the condition in the Bohr-Sommerfeld theory that the action integral of a system must be an integral multiple of Planck's constant.

QUANTUM INTERFEROMETRY: The study of interference phenomena in which the interference is caused by the quantum mechanical wave nature of particles such as electrons and neutrons. The standard quantum limit results from radiation pressure noise and photon shot noise. It is believed to be the fundamental limit for interferometric length measurements.

QUANTUM JUMP (QUANTUM LEAP): The change in the quantum state of a system such as an atom or molecule from one state to another. This causes the emission of electromagnetic radiation, including that of light, which occurs in the form of quantised units called **photons**.

QUANTUM MECHANICS: 1. Also known as **quantum physics** or **quantum theory**, it is a branch of physics providing a mathematical description for wave and particle behaviour at the subatomic level. It differs from classical mechanics, which describes the behaviour of macro molecules whilst quantum mechanics deals with particles which are smaller than an atom such as electrons, protons, neutrons, nuclei, etc. 2. Also known as **molecular mechanics**, it is a hybrid procedure for the treatment of large molecular systems. A crucial part of the system, for example, the chromophore and its surroundings is treated explicitly by quantum mechanical (QM) techniques, whereas the rest of the system is approximated by a classical or molecular mechanics (MM) force field treatment.

QUANTUM MODELS OF SPECIATION: Models of evolution that hold that speciation sometimes occurs rapidly as well as over long periods, as the classical theory proposed.

QUANTUM NUMBER: A set of integer or half integer values needed to characterise the physical properties and energy states of atomic or subatomic particles. The quantum number assigned to an electron in an atom serves to define such a system thereby indicating the possible states the system may have. For example, principal quantum, n defines the electron's main energy level and takes values $n = 1, 2, 3, 4...$

QUANTUM OPTICS: The study of the aspect of light and its interactions with matter that rely on quantum mechanics. Quantum Optics is applied in quantum electronics, photonics and laser study.

QUANTUM PHASE TRANSITION (QPT): A phase transition that occurs at absolute zero temperature when some parameters such as a magnetic field or pressure are changed. In contrast to ordinary thermal phase transitions, quantum phase transitions are associated with quantum fluctuations.

QUANTUM SIMULATION: The mathematical modelling of systems of large numbers of atoms or molecules by computer studies of relatively small clusters. It helps to obtain information about solids and liquids by quantum simulation and to study surface properties and reactions.

QUANTUM SPINS LIQUID: A quantum-mechanical system in which a large number of electron spins are coupled together to give

a distinct quantum system. Quantum spin liquid can be used to investigate aspects of the many-body problem in quantum mechanics and have potentially important applications in technology.

QUANTUM STATE: The state of a quantised system as described by its quantum numbers. The spin of an isolated electron can take on one of only two values. There are no other quantum states available for the electron and no intermediate values, since spin is quantised. The quantum state is sometimes described by a set of quantum numbers that pick out the appropriate values for describing the state. For example, the state of a hydrogen atom being described by the four quantum numbers as n, l, m, s . In the grouped state they have values 1, 0, 0 and $1/2$ respectively.

QUANTUM STATISTICS: A statistical description of systems of particles that obey the rules of quantum mechanics rather than classical mechanics.

QUANTUM TELEPORTATION (ENTANGLEMENT-ASSISTED TELEPORTATION): The transfer of quantum information from one quantum system to another across space. Quantum teleportation has been performed both for photons and for systems with electrons. It is used for quantum communication and computation.

QUANTUM THEORY: A physical theory which describes the motion of material particles and energy states based on the assumptions that energy is subdivided into discrete amounts and that matter possesses wave properties.

QUANTUM THEORY OF RADIATION: The theory of the emission and absorption of electromagnetic radiation based on Planck's law. Since it is a quantum theory the processes of emission and absorption are described in terms of the creation and disappearance of photons, with photons being created in emission processes and disappearing in absorption. Absorption occurs when electromagnetic radiation that impinges on a quantum mechanical system, such as an atom or molecules, induces a transition from the ground state of the system to an excited state by virtue of a photon of the radiation being absorbed. Similarly, emission of a photon occurs when an atom returns to the ground state from an excited state (spontaneous emission occurs). It is also possible for an atom in an excited state to emit a photon by induced emission, a process that is used in lasers.

QUANTUM TURING MACHINE (QTM; UNIVERSAL QUANTUM COMPUTER): A mathematical concept of a theoretical computing device, proposed by Alan M. Turing (1912-1954) in 1936. Any quantum algorithm can be expressed formally as a particular quantum Turing machine. This machine, which could theoretically perform any mathematical calculation, was important in the development of the digital computer.

QUANTUM WELL: A potential well in which electrons in a three-dimensional system are confined to a plane or two dimensions.

QUANTUM YIELD: 1. In a photochemical reaction, the number of defined events occurring per photon absorbed by the system expressed as the number of reactant molecules decomposed divided by the number of moles of photons absorbed:

$$\frac{\text{no. of reactant molecules decomposed}}{\text{no. of photons absorbed}}$$
 2. A measure of the efficiency with which absorbed light produces some effect. It is expressed by

the equation:
$$\frac{\text{no. of photons emitted}}{\text{no. of photons absorbed}}$$

QUANTUM ZENO EFFECT: A phenomenon in which constant observation of an unstable particle prevents it from decaying.

QUARANTINE: 1. The isolation of people, animals, plants or pest-infested materials suspected of having an infectious disease, until they have been cleared of the infection. 2. The removal of an undesired file

such as a virus-infected file or spyware to a folder that is not easily accessible by regular file management utilities.

QUARK: One of a set of elementary particles or fundamental particles not known to be made up of smaller particles (electrons, protons, neutrons, etc.) and a fundamental constituent of matter with an electric charge equal to one-third or two-thirds that of the electron. Quarks are the constituents of baryons and mesons. The six types of quarks known occurring in three pairs are up and down, strange and charm and top and bottom. Only the up and down quarks are needed to make protons and neutrons; the others occur in heavier, unstable particles.

QUARK CONFINEMENT: The hypothesis that constituent quarks in a group can never be removed from the parent hadrons, even though the interactions between them are relatively weak. Quarks can only be studied or observed at a hadron level because of this phenomenon.

QUARK STAR: A hypothetical star composed of free quarks with a density intermediate between that of a neutron star and a black hole.

QUARK-GLUON PLASMA: A predicted state of matter containing quarks and gluons which are not confined into baryons and mesons but exist as hot plasma. This state is thought to have existed in the early universe for about 1 to 10^{-3} seconds after the big bang, when the universe had cooled to a low temperature to permit quark confinement into hadrons.

QUART: A unit of fluid measurement which is equal to 2 pints. One imperial quart is equal to 1.13652 litres and one U.S. quart is equal to 0.946353 litres.

QUARTER INCH CARTRIDGE (QIC): A magnetic tape data storage tape with a capacity of 25 GB.

QUARTER-WAVE PLATE: A thin sheet of mica or other doubly refracting crystal material of such thickness as to introduce a phase difference of one quarter-cycle between the ordinary and the extraordinary components of light passing through. Such a plate converts circularly polarised light into plane-polarised light.

QUARTIC: A polynomial function or a polynomial equation, whose highest order of degree or exponent is four. An example is $f(x) = 2x^4 - 7x^3 + x - 3$.

QUARTILE: 1. Any of the three values that divides a frequency distribution into four equal parts such that each contains a quarter of the sampled population. The quartiles positions are given as follows: first quartile (Q_1) = $\frac{1}{4}\sum f$; second quartile (Q_2) = $\frac{1}{2}\sum f$; and third quartile (Q_3) = $\frac{3}{4}\sum f$. 2. The value below which one-quarter of data lies and the value above which three quarters of the set of data lies. 3. The astrological aspect of planets which are distant from each other by 90° or one-fourth of the zodiac.

QUARTILE DEVIATION (Q.D.; SEMI-INTERQUARTILE RANGE): A measurement of spread or variability defined as half of

the difference between the first and third quartiles: $Q.D. = \frac{Q_3 - Q_1}{2}$. $Q_3 - Q_1$ is known as the interquartile range.

QUARTZ: The second (after feldspar) most abundant and common mineral on the surface of the Earth, forming about 12% of the crust by volume. It consists of a form of crystalline silicon dioxide (SiO_2) having a hardness of 7 on the Mohs' scale and a specific gravity of 2.65. Well-formed crystals of quartz are six-sided prisms terminating in six-sided pyramids. Rock crystal is the most common quartz and is colourless and transparent; but there are coloured varieties, some of which are used as gemstones. Examples are amethyst, citrine quartz (yellow), rose quartz (pink), milk quartz (white), smoky quartz (grey-brown),

chalcedony, agate and jasper. Quartz occurs in igneous rocks and especially in granites, granodiorites, pegmatites and rhyolites. It also occurs in metamorphic rocks such as phyllites, quartzites, schists, granulites and eclogites. It is also found in sedimentary rocks such as sandstone and limestone. It is used in optical instruments and in oscillators, glass, glaze and abrasives.

QUARTZ CLOCK: A clock in which the time-keeping is regulated by a quartz crystal based on its piezoelectric properties that vibrates at a fixed frequency in an electronic oscillator.

QUARTZ IODINE LAMP: An electric lamp having a tungsten filament with a bulb made of quartz glass filled with iodine vapour. It is used in automobile headlights and movie projectors.

QUARTZITE: A common metamorphic rock consisting of pure quartz sandstone that has recrystallised under increasing heat and pressure. They are mostly white, light-grey, yellowish or light brown; but also blue, green, purple or black. Some quartzites originate from sedimentary rocks, as a result of cementation of quartz sandstone by groundwater solutions containing pure quartz.

QUASAR: An acronym for **quasi-stellar radio source**. It is an energetic and an extremely remote compact region in the centre of a massive galaxy surrounding the central super massive black hole, usually with a large redshift source of electromagnetic energy, such as radio waves and visible light that emits huge amounts of energy. It is the most luminous object in the universe. **Quasi-stellar object (QSO)** is a name used by some astronomers for quasars, because most quasars are not strong radio-wave emitters.

QUASI-: Sharing some important property with a given class of objects, such as quasi-analytic functions and quasi-linear equations.

QUASI-ATOM: A structure of atomic or sub-atomic particles, in which the nuclei of two colliding atoms appear to have become bound together for a brief period. This can occur when the nuclei of the two set of particles colliding become much closer to each other than they are to their constituent electrons.

QUASI-CLASSICAL TRAJECTORY METHOD (QCT METHOD): A procedure for calculating trajectories in which the quantisation of the reactants is taken into account, but in which the course of the reaction is treated classically.

QUASI-CONCAVE: In mathematics, a quasi-concave function is a function having convex upper level sets; all sets of the form $\{x : f(x) \geq r\}$ are convex for all r .

QUASI-CONVEX: In mathematics, a quasi-convex is a function having convex lower level sets; all sets of the form $\{x : f(x) \leq r\}$ are concave for all r , as holds for any convex function.

QUASI-ENANTIOMERS: Constitutionally different yet closely related chemical species MX and MY having the opposite chirality sense of the large common chiral moiety M. For example, (R)-2-bromobutane is a quasi-enantiomer of (S)-2-chlorobutane.

QUASI-EQUILIBRIUM: In conventional transition-state theory, it is assumed that activated complexes are formed in a state of equilibrium with the reactants. They are not in classical equilibrium with the reactants; if they were, addition of more activated complexes to the system would cause the equilibrium to shift in favour of the reactants. This would not occur for an activated complex and the term quasi-equilibrium is used to denote this special type of equilibrium.

QUASI-EXPERIMENT: A term that is used for studies that resemble experiments but are weak on some of the characteristics, particularly that allocation of subjects to groups not under the investigator's control.

QUASI-GROUP: In mathematics, a groupoid in which every element has both a unique left inverse and a unique right inverse, which

need not be equal unless the associative law holds. If a quasi-group is commutative, each element has at least one inverse, but it need not be unique.

QUASI-LINEAR: Of an n^{th} order ordinary differential equation, such that it can be written in the form $y^{(n)} = f(x, y', \dots, y^{(n-1)})$. Quasi-linear ordinary differential equations of any order can be written as a system of first order ordinary differential equations.

QUASI-MOLECULAR ION: A protonated molecule or an ion formed from a molecular ion by the loss of a hydrogen atom.

QUASI-NEWTON METHOD: In mathematics, any of a class of methods for solving non-linear equations or unconstrained optimisation problems, based on Newton's method but using rough estimates of the Hessians or Jacobians involved in order to reduce computational cost. These estimates are often recursively estimated and are refined during the calculation. In quasi-Newton minimisation procedures, the approximate Hessian H is usually given a rank one or rank two update and a line search method is then used in the direction of $-H^{-1} \nabla f(x)$ to produce the next estimate.

QUASI-REGULAR POLYHEDRON: A polyhedron that consists of two sets of regular polygons, m -sided and n -sided respectively and is constructed so that each polygon in one set is surrounded by members of the other set. There are three convex quasiregular solids: the **cuboctahedron** ($m = 3, n = 4$), the **icosidodecahedron** ($m = 3, n = 5$) and the **octahedron** ($m = n = 3$). Among the nonconvex polyhedra are two examples of type (m, n, m, n) : the **dodecadodecahedron** ($m = 5, n = \frac{5}{2}$); and the **great**

icosidodecahedron ($m = 3, n = \frac{5}{2}$), which can be made by truncating the Kepler-Poinsot polyhedra at their edge midpoints. There are also three nonconvex examples of type (m, n, m, n, m, n) : the **small triambic icosidodecahedron** ($m = 3, n = \frac{5}{2}$), the **triambic dodecadodecahedron** ($m = \frac{5}{3}, n = 5$) and the **great triambic icosidodecahedron** ($m = 3, n = 5$).

QUASI-SINGLE-STRAND POLYMER: A regular linear polymer that can be described by a preferred constitutional repeating unit in which only one terminal constituent subunit is connected through a single atom to the other identical constitutional repeating units or to an end group.

QUASI-STELLAR OBJECT (QSO): A name used by some astronomers for quasars, because most quasars are not strong radio-wave emitters. **Quasar** is an acronym for **quasi-stellar radio source**, but it has been found out that not all quasars discovered are strong radio-wave emitters.

QUASI-STELLAR RADIO SOURCE (QUASAR; QSS): An energetic and an extremely remote compact region in the centre of a massive galaxy surrounding the central super massive black hole, usually with a large redshift source of electromagnetic energy such as radio waves and visible light that emits huge amounts of energy. It is the most luminous object in the universe.

QUASI-TAUTOLOGY: In logic, a tautological consequence of instances of identity and equality axioms in a theory.

QUASICRYSTAL: A solid with conventional crystalline properties that form patterns that fill all the space but lack translational symmetry.

QUASIPARTICLE: A particle-like unit formed when electrons in very strong magnetic fields react with each other.

QUASIPERIODIC: A form of motion that is regular but never exactly repeating. Quasiperiodic motion is always composed of multiple but simpler periodic motions. In the general case, for motion that is the

sum of simpler periodic motions, if there exists a length of time that evenly divides the frequencies of the underlying motions, then the composite motion will also be periodic. However, if no such length of time exists, then the motion will be quasiperiodic.

QUATERNARY: 1. Having four variables. 2. Containing atoms bonded to four other atoms. 3. The current period of geologic time divided into the Pleistocene Epoch and Holocene that follows the Neogene Period starting from about 1.6 million years ago and characterised by the appearance and dominance of humans.

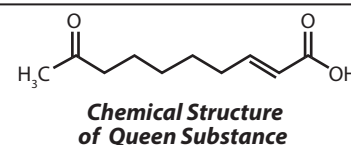
QUATERNARY AMMONIUM COMPOUND (QUATS): A white crystalline solid nitrogen compound, soluble in water, which completely dissociates in solution. It has the formula, $R_4N^+X^-$, where R is a long chain alkyl group. It has a nitrogen atom joined to four organic radicals and one acid radical and regarded as a derivative of ammonium. They are used as solvents, antiseptics, emulsifying agents and disinfectants.

QUATERNARY STRUCTURE: The structure formed in some proteins, by the noncovalent interaction of two or more macromolecules. Examples are interaction of four globin protein molecules to make haemoglobin and histones interacting with DNA to make a nucleosome.

QUATERNION: A type of higher complex number defined by $Q = a + bi + cj + dk$, where a, b, c and d are real numbers and i, j and k are unit vectors in the x, y and z directions respectively. Its symbol is Q . It was first suggested by Sir William R. Hamilton in 1843.

QUEEN: A large, fully developed fertile female that lays eggs in a colony of social insects such as bees, ants and termites.

QUEEN SUBSTANCE: A pheromone, trans-9-keto-2-decenoic acid, secreted by the mandibular salivary glands of queen honey bees and consumed by the workers that suppresses the development of ovaries in the worker bees preventing them from egg-laying.



QUELEA: A brownish bird of the genus *Quelea*, of the weaverbird family and native to Africa that live and breed in huge flocks. The Red-billed Quelea, *Quelea quelea*, is the world's most abundant wild bird species, with an estimated adult breeding population of 1.5 billion pairs. Red-billed Quelea are small birds and average-sized weaverfinches, which grow to 11–13 cm long and a weight of 15 to 20 g. The average quelea bird eats around 10 grams of grain per day and is extremely destructive to grain crops.

QUENCHER: A molecular entity that deactivates (quenches) an excited state of another molecular entity, either by energy transfer, electron transfer or by a chemical mechanism.

QUENCHING: The rapid cooling of a metal by immersing it in a bath of liquid in order to obtain certain material properties such as reducing crystallinity leading to increased toughness of both alloys and plastics. Some nonferrous metals such as copper are quenched to make them softer.

QUERY: In computing, it is a request for specific data or information from a database.

QUERY BY EXAMPLE (QBE): A search method based on selecting text string or a form of document that is passed on to the engine, which then formulates the actual query. As such, it is easier to learn the ordinary query language.

QUERY LANGUAGE: A computer language such as SQL and MySQL used in searching databases for purposes of reporting.

QUERIES PER SECOND (QPS): A measure of how much traffic a server is handling at any given time.

QUEUE: In computing, a list of data items, commands or instructions that are stored and waiting to be processed or served in a defined order.

QUEUEING THEORY: A mathematical theory which is concerned with the study of various factors relating to queues such as the distributions of arrivals and the average time spent in queues.

QUICK CLAY: Any sensitive water-saturated clay having the tendency to change from a relatively stiff condition to a fluid mass when it is disturbed.

QUICK FORMAT: This is a method of removing all data or files from a disk or a partition so that it can be written on. It erases files from the partition but does not scan the disk for bad sectors. It takes a very short time to complete. This option can be used if the hard disk has been previously formatted and you are sure that the disk is not damaged.

QUICK LAUNCH: A section on a Microsoft Taskbar that enables a user the ability to launch their programs.

QUICKLIME (CALCIUM OXIDE): A colourless, cubic crystalline or white amorphous substance with the chemical formula CaO , density 3.35 g/cm^3 , melting point 2572°C (2845 K) and boiling point 2850°C (3123 K). It is prepared by heating calcium carbonate such as limestone in a special lime kiln to about 500°C (773 K) to 600°C (873 K), decomposing it into the oxide and carbon dioxide. It is used as a refractory, as a flux, in manufacturing steel and paper, in glassmaking, in waste treatment, in insecticides and as an industrial alkali.

QUICKSAND: Sand saturated with water to form a sand-water suspension possessing the characteristics of a liquid. Quicksands may form at river mouths or on sand flats and are dangerous as they appear identical to adjacent sand.

QUICKSILVER: 1. Also known as **mercury**, it is a chemical element with the symbol Hg , atomic number 80, freezing point -38.83°C (234.32 K) and boiling point of 356.73°C (629.88 K). It is a heavy, silvery d-block element and the only metal that is liquid at standard conditions for temperature and pressure. Mercury occurs in deposits throughout the world mostly as cinnabar (mercuric sulfide). It is used in thermometers, barometers, manometers, sphygmomanometers, float valves, some electrical switches and other scientific apparatus. Mercury and most of its compounds are extremely toxic and must be handled with care. 2. A fast and free Mac OS X computer utility software productivity application that allows users to use the keyboard to rapidly perform tasks such as launching applications, manipulating files or sending e-mail. It has three panes, into which the user can enter an object, an action and an optional attribute.

QUICKTIME (QUICK TIME): A software that allows a computer user to play audio and video files. It was developed by Apple and first released on December 2, 1991. QuickTime is available for Apple and IBM compatible users running a compatible operating system such as Microsoft Windows or any Macintosh operating system. With IBM compatible computers, QuickTime is commonly used to play .MOV extension files.

QUILL FEATHER: Any of the main wings or tail feathers of a bird used for flight and sexual display.

QUIESCENT CENTRE (QUIESCENT ZONE): A central region, which is usually hemispherical, located within the zone of cell division in plant roots, containing meristematic cells that rarely divide or divide very slowly.

QUIESCENT CURRENT: Current that is consumed by a battery when the application it powers is in standby or hibernation mode.

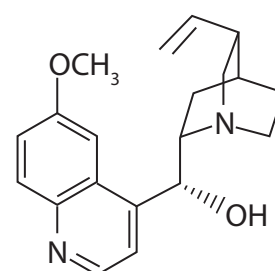
QUINARY: 1. Occurring in fives. 2. In mathematics, a base of five.

QUINCUNCIAL: Having a five-ranked leaf arrangement.

QUINHYDRONE ELECTRODE: A half-cell consisting of a platinum electrode in contact with quinhydrone crystals and a water-based solution. It is slightly soluble in water and dissolves to form a mixture of two substances, quinone and hydroquinone, which are present at equal concentration and can easily be oxidised or reduced to the other. It is used to measure the hydrogen ion concentration (pH) of a solution in a chemical experiment.

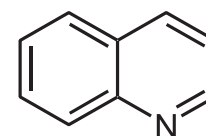
QUINHYDRONES: Molecular complexes of one equivalent amount of a quinone with one equivalent amount of the corresponding hydroquinone.

QUININE: A white, bitter crystalline alkaloid with the chemical formula $\text{C}_{20}\text{H}_{24}\text{N}_2\text{O}_2 \cdot 3\text{H}_2\text{O}$. It is derived from certain cinchona barks and used in medicine to treat malaria. It is also used as a flavouring agent in some carbonated beverages, including tonic water.



Chemical Structure of Quinine

QUINOLINE (1-BENZAZINE; 1-AZANAPHTHALENE; BENZO[b]PYRIDINE): A hot water-soluble, colourless, hygroscopic, oily aromatic nitrogen compound liquid with a heterocyclic base having the chemical formula $\text{C}_9\text{H}_7\text{N}$. It can be found in bone oil and coal-tar and used in making dyes and drugs.



Chemical Structure of Quinoline

QUINONE: A group of cyclic aromatic diketone compounds comprising a six-membered unsaturated ring to which two oxygen atoms are bonded as carbonyl groups. It is derived from aromatic compound such as benzene or naphthalene, where $(-\text{CH}=)$ is replaced by $-\text{C}(=\text{O})-$.

QUINONIMINES (QUINONE IMINES): Imines derived from quinones by replacement of one or both oxygens by $=\text{NR}$. The term may include both types unless the infix mono or di is included, for example, o-benzoquinone diimine.

QUINONOXIMES: Mono- or di-oximes of quinones; quinone monooximes are tautomeric with nitrosophenols.

QUINQUECOSTATE: Having five ribs.

QUINQUEFARIOUS: Arranged in five ranks.

QUINQUEFOLIATE: Having five leaves or leaflets.

QUINQUEJUGATE: Arranged in five pairs.

QUINQUELOCULAR: Having five cells or locules.

QUINQUENERVED: Having five main nerves.

QUINQUEPARTITE: 1. Divided into or consisting of five parts. 2. Composed of five participants or groups of participants.

QUINTAL: A unit of mass sometimes used to measure bulk agricultural commodities. It is equal to 100 kg.

QUINTESSENCE: A hypothetical form of energy postulated as an explanation of observations of an accelerating universe.

QUINTIC: A polynomial function or equation of the fifth order or having a property of five. An example is $f(x) = 3x^5 + 6x^4 - 2x^2 + x - 1$.

QUINTILES: The set of four variate values that divide a frequency distribution or a probability distribution into five equal parts.

QUOD ERAT DEMONSTRANTUM (QED): A Latin phrase, which means that "which is to be proved or shown". It is put at the end of a proof to signify that the proof has been completed.

QUORA: A Q&A (question-and-answer site) social networking website launched in June, 2009 in which users can collaborate by editing questions and suggesting edits to answers that have been submitted by other users. It was co-founded by two former Facebook executives, Charlie Cheever and Adam D'Angelo.

QUORUM SENSING: A type of decision-making process used in communication and coordination among populations of bacteria in which the individual bacteria secretes and detects substances known as autoinducers, which accumulate in the environment depending on the density of the population. When a certain threshold concentration is reached, the autoinducer triggers a special response, regulating many bacterial activities such as bioluminescence, sporulation, virulence and biofilm formation. Quorum sensing is also used in applications for computing and robotics.

QUOTA SAMPLING: A survey technique in which there is no random selection of units, but the sample is first segmented into mutually exclusive sub-groups such as the percentage of people who come from various identifiable subpopulations, for example, men over 50, women under 30, employed/unemployed men, etc.

QUOTIENT: 1. The result of dividing one number, quantity or expression by another, called the divisor. For example, the quotient of twelve divided by 3: $12 \div 3$ is 4. 2. The ratio of two numbers or quantities. Also, the integral part of such a result.

QUOTIENT GROUP (FACTOR GROUP): The group, denoted G/H , whose members are the cosets of a normal subgroup H in a group G , with the product of two cosets defined as the coset containing the product in the same order of arbitrary representatives of the given cosets.

QUOTIENT OF POWERS PROPERTY: It states that when two numbers with the same base and different exponents are divided, the quotient equals the same base number raised to the difference of exponents, for example, $10^5 \div 10^2 = 10^{5-2} = 10^3$.

QUOTIENT RING: 1. Also known as **factor ring** or **residue class ring**, it is the ring, denoted R/K , whose members are the cosets of an ideal K in a ring R ; these cosets are called residue classes and are equivalence classes whose members differ by a member of K . Since K is an ideal, the sums and products of elements of the factor ring are unique and the cosets of a sum or product of ring elements equals

the sum or product (in the same order) of their individual cosets. K is the zero of the factor ring and it has a unit element if R has. 2. An over-ring, Q , of a given ring, R , with some regular elements, that consists of elements of the form ab^{-1} (in a right quotient ring) or $b^{-1}a$ (in a left quotient ring) where a is any element and b is a regular element of R and the inverses of all regular elements of R are in Q . Any semi-prime right Noetherian ring has a semi-simple right quotient ring and any integral domain has a quotient ring that is a field, the field of fractions.

QUOTIENT RULE: In mathematics, the rule for differentiating the quotient of two for which derivatives exist, given by

$$\frac{d\left(\frac{f}{g}\right)}{dx} = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2}. \text{ In the equation, } f \text{ and } g \text{ are functions.}$$

QUOTIENT SPACE (FACTOR SPACE): A space obtained from another by identification of points that are equivalent to one another in some equivalence relation. For example, the factor group G/H of a group G by a normal subgroup H is the set of cosets of H in G and the factor ring R/K of a ring R by an ideal K is the set of cosets of K in R .

QUOTIENT TOPOLOGY: In mathematics, the quotient topology on a set, Y , with respect to a topological space, X and a mapping, f , from X onto Y , is the finest topology with respect to which f is continuous.

QVGA (QUARTER VIDEO GRAPHICS ARRAY): A small-screen computer display in which the resolution is 320 pixels horizontally by 240 pixels vertically. QVGA displays are used in small computers and consumer electronic devices such as mobile phones, personal digital assistants (PDAs), digital cameras, camcorders, handheld game consoles, and portable MP3.

QWERTY KEYBOARD: An alphanumeric keyboard used on Standard English computers and typewriters, named after the first six letters on the top rank of letters.

QZONE: A Chinese-based social networking website created in 2005 that is popular with young teenagers and rural users. It allows users to write blogs, keep diaries, send photos, listen to music, and watch videos. It is one of the top four social networking platforms in China as of 2018. The others are: RenRen (formerly Xiaonei), Kaixin001, and 51.com. As at August 2018, it has 563,000,000 monthly active users.

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r: 1. Abbreviation for radius. 2. It is the function, $r(n)$, that counts the number of representations of a natural number, n , as a sum of two integral squares, counting representations as different even when they only differ in order or sign. This is equal to $4[d_1(n) - d_3(n)]$, where $d_i(n)$ is the sum of the divisors of n that are congruent to i modulo 4 and is multiplicative. For example, $r(5) = 8$; in general $r(p) = 8$ for any prime that is congruent to 1 modulo 4. 3. The function, $r_k(n)$, that counts the number of representations of a natural number, n , as a sum of k integral squares, counting representations as different even when they only differ in order or sign.

R: The symbol for an unspecified relation, usually written either in prefix notation as R_{xy} or xRy for binary relations in infix notation to mean x has relation R to y .

R: 1. The symbol for the set of real numbers. 2. Symbol for molar gas constant.

R GROUP: The specific identity side chain of an amino acid. There are 20 amino acids present in proteins, each with a specific R group or side chain. It gives each amino acid proteins specific characteristics, including size, polarity and pH. R group is also used to denote virtually any organic substituent.

R PLASMID (R-PLASMID): Abbreviation for **resistance plasmid**, a bacterial plasmid carrying genes that confer resistance to certain antibiotics.

R-PROCESS (RAPID PROCESS): The nucleosynthesis process in which a nucleus captures a large number of neutrons in rapid succession. The r-process can only occur when there are many neutrons available, as in a supernova explosion. It is thought that many elements that are heavier than iron, such as gold, were formed in this way.

R-PROGRAMMING LANGUAGE: An open source scripting language for predictive analytics and data visualisation, particularly useful in statistical analysis.

R-SELECTED SPECIES (R-STRATEGIST SPECIES): Species whose populations increase rapidly, often exponentially, quickly filling available environments. Some characteristics of r -selected species include the following: shorter gestation periods; numerous smaller offspring, early maturation and therefore, require little or no parental care; shorter life spans; little competition for resources; and reproduction at a relatively young age but many offspring die before they reach reproductive age. They thrive in disturbed habitats, such as freshly burned grasslands or forests characterised by canopies that open abruptly. Examples of r -selected species include pest organisms, such as rodents, insects and weeds.

R-SELECTION: A type of selection that favours organisms with a high biotic potential (optimal ability to survive and reproduce successfully). Traits that are thought to be characteristic of r -selection include high fecundity, small body size, early maturity onset, short generation time, and the ability to disperse offspring widely. Organisms that are selected are able to colonise a habitat rapidly before other organisms are established and begin to compete. Examples of organisms'

r -selection are bacteria and diatoms, insects, weeds, semelparous cephalopods and rodents.

RAABE'S TEST (RAABE'S CONVERGENCE TEST): In mathematics, a test for the convergence of a series: if $a_n < 0$, and $\frac{a_{n+1}}{a_n} < 1 - \frac{A}{n}$ for some constant A greater than 1, and sufficiently large n , then $\sum a_n$ converges; for example, the hypergeometric series converges by this test.

RABBIT: 1. A small common herbivorous burrowing mammal of the family Leporidae, which also includes hares and pikas. It has long ears, soft fur, a short tail and large hind legs and feet adapted for running or jumping. Unlike hares, rabbits are born blind, hairless, and helpless. Rabbits are commonly raised for meat, for their fur and sometimes as pets.

Hares belong to the genus *Lepus*. They are similar to rabbits but have longer ears and legs and give birth to active, furred young with eyes open. 2. A term used in nuclear analytical chemistry to refer to a small container propelled pneumatically or hydraulically through a tube leading from the laboratory to a location in a nuclear reactor or other device where irradiation of a sample can take place.

RABBIT HUTCH: A container with a wire front in which a rabbit is kept.

RABI MODEL: A quantum mechanical model describing the interaction between a two-level system and an electromagnetic field. The model was originally proposed by the Galician-born American physicist Isidor Rabi (1898–1988) in the context of NMR, with the two levels being spin-up and spin-down and the field being an external magnetic field.

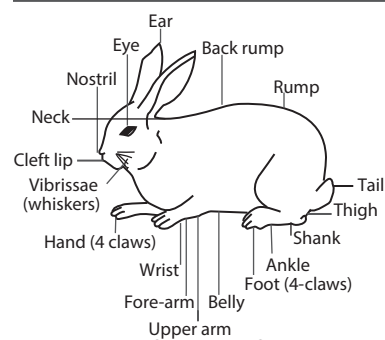
RABIES: An acute viral disease that causes inflammation of the brain and affects the central nervous systems of most warm-blooded animals. It is transmitted in the saliva of an infected animal most commonly by a bite. The rabies virus travels to the brain by following the peripheral nerves. It causes convulsions, inability to move, and strange behaviour. The main carriers include dogs, cats, foxes and bats. If not treated in time (within a day or two) with a serum containing antibodies and then a series of vaccinations, rabies in humans is almost always fatal.

RACE: A population of interbreeding species that develops distinct characteristics differing from other populations of the same species, especially as caused by geographical isolation.

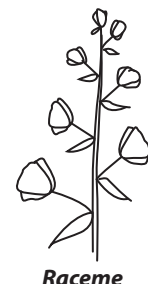
RACEMATE: An equimolar mixture of a pair of enantiomers. It does not exhibit optical activity. The chemical name or formula of a racemate is distinguished from those of the enantiomers by the prefix ($\pm$)- or *rac*- (or *racem*-) or by the symbols *RS* and *SR*.

RACEME: A type of inflorescence in which the main flower stalk is unbranched and elongated and bears pedicels. The lower older flowers have longer stalk flowers that are borne towards the base and the top new flowers are produced as the shoot grows. An example is *Crotalaria retusa* (rattlepod or rattlebox).

RACEMIC COMPOUND: A crystalline racemate in which the two enantiomers are present in equal amounts in a well defined arrangement within the lattice of a homogeneous crystalline addition compound.



External features of a rabbit



Raceme